

Deep Ground Repository

The Conceptual Model and Analysis of a Deep Underground Storage for Radioactive Waste

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Abstract

This study investigates the impact of thermal and mechanical processes in deep geological repositories (DGRs) during the construction and early closure phases. The analysis focuses on pore water pressure, volumetric strain, temperature changes, stress distribution, and deformation in two cases of underground tunnels with varying spacing and liner thicknesses. In the first case, the research found out the deformation due to the constructions of the tunnel is minimal due to the small size of the tunnel, on the other hand the thermal coupling play a major role especially in the long term as shown by significant increase of porewater pressure from 760 kPa during construction to 5,000 kPa near the tunnels in the early closure phase and double the amount during late closure phase.

The volumetric strain exhibits negative values during excavation, indicating shrinkage, but transitions to positive values due to thermal expansion during initial heating, peaking at 0.02 near the tunnels. In the second case, featuring 100-meter spacing and a thinner 0.4-meter liner, the second case has reduced interference which led to lower pore water pressures and volumetric strains compared to the first case and less deformation during the construction phase. The results indicate that increased spacing between tunnels minimizes heat buildup, thereby reducing thermal stress and deformation in surrounding formations while also has reduce interferenc of redistribution of stress during the construction phase. This research shows that the changes due to introduction of heat from the radioactive play a bigger role for the changes that happen in the facility and the host formations rather than the construction itself. Recommendations obtained from this research results emphasize using reinforced concrete for liners and maintaining adequate tunnel spacing to enhance the stability and integrity of DGRs over the long term.

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Nomenclature

Abbreviations

Abbreviation	Definition
DGR	Deep Geological Repository
EDZ	Excavation Damage Zone
LLW	Low-Level Waste
ILW	Intermediate-Level Waste
HLW	High-Level Waste
SF	Spent Fuel

Symbols

Symbol	Definition	Unit
T	Temperature	[°]
ρ	Density	[kg/m ³]
u	Displacement	[m]
σ	Stress	[Pa]
K_0	Coefficient Earth Pressures at Rest	
σ'_1	Effective Principal Stress	[Pa]
γ	Unit Weight	[kN/m ³]

1

Introduction

Radioactive waste are type of waste that contains radioactive material. This radioactive are caused by the decaying mass of the object thus we as human are trying to solve and research in disposing radioactive in safe space that are outside of human activity and withstand time.

The concept of engineered geologic disposal has been developed for the safe long-term management of long-lived radioactive waste. This involves the emplacement of such waste in deep underground repositories that provide for the secure and safe isolation of the waste and, consequently, the protection of humans and the environment. The aforementioned is chosen as placing disposal underground far away from any human activity are way to ensure the radioactive waste does not go interact with the surface where human activity lies.

Several studies are already conducted and several URL(Underground Research Laboratory) are constructed e.g.(HADES, Switzerland). The aim for this URL is to identify and mimic the physical process that would occur during the the construction, emplacement and post closure of a deep underground radioactive waste. It is indicated that the creating the tunnel would introduce a redistribution of stress due to underground excavation, changing parameter of underground host formation. Thermal coupling due to the heat emitted by radioactive waste and also the hydraulic condition that would substantially changes due to construction and also to the emitted heat.

1.1. Assignment Goals

This assignment goals is:

1. Identify the design questions at the di erent time periods (construction, emplacement, early post-closure, very long-term).
2. Create conceptual models of the processes affecting the disposal facility over the different time periods.
3. Create a numerical model of a geological disposal facility for the early post-closure phase (thermal phase). Perform a 2D analysis of the problem using the software Plaxis.
4. Analyse the stress, water pressure and temperature fields for the base case and different values of the design parameters and optimise the design of a disposal facility based on the parameter-s/geometry sensitivity analysis.

2

Literature Review

This chapter would discuss the conceptual model, theory based on a literature review and the definition of problem that are dealt in this assignment.

2.1. Radioactive Waste

Radioactive waste are type of waste that contains radioactive material. Radioactive waste are but not limited to waste that is produced by nuclear reactor, rare-earth mining, radioactive medicines and instruments and nuclear weapons [2].

Radioactive waste is broadly classified into 3 categories:

1. Low-level waste (LLW), such as paper, rags, tools, clothing, which contain small amounts of mostly short-lived radioactivity;
2. Intermediate-level waste (ILW), which contains higher amounts of radioactivity and requires some shielding;
3. High-level waste (HLW), which is highly radioactive and hot due to decay heat, thus requiring cooling and shielding.

2.2. Deep Geological Repository

A deep geological repository is a way of storing hazardous or radioactive waste within a stable geologic environment, typically 200–1,000 m below the surface[1].It entails a combination of waste form, waste package, engineered seals and geology that is suited to provide a high level of long-term isolation and containment without future maintenance. The methods of sotring the radioactive material in the DGR are also improved year after year. The recent designs was to implement a multiple barrier on the radioactive called supercontainer. This is done by applying multiple layer of barrier in the radioactive waste to ensure no leak are apparent in the waste.

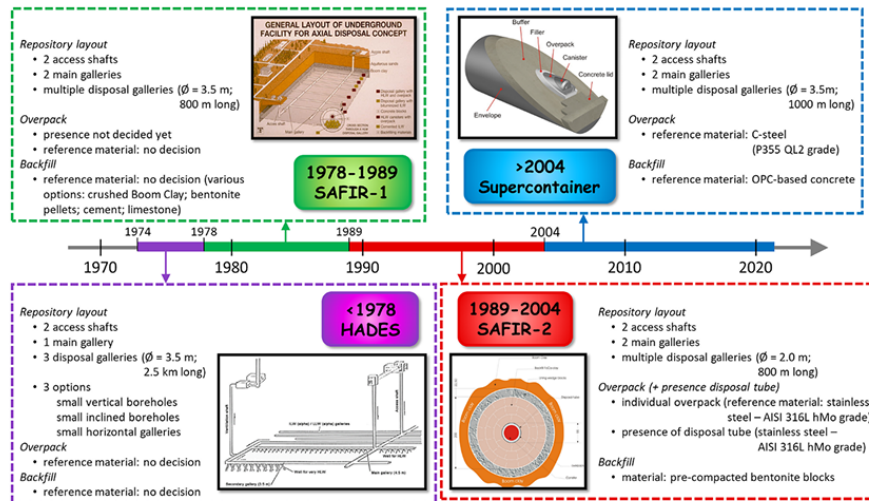


Figure 2.1: DGR Method

2.3. Literature Review

Undergoing research upon understanding the behaviour of the deep geological repository due to thermal waste has been extensively conducted in several sites. Notable projects include HADES URL, Grimsel Site test and Beishan URL.

The HADES URL is constructed in Mol, Belgium around a 200m depth in a boom clay formation. This place is chosen as the surrounding formation has small permeability that reached around $10^{-12} m/s$. During the construction the research investigates how the deformation would take places around the HADES URL and they found out that the deformation around the tunnel aren't as much as they expected it to be.

The HADES URL conducted PRACLAY Heater Test. This test is done to simulate the heat that would be generated by SF and HLW ad the disposal. The 3 main focus of this PRACLAY heater test is how temperature affect the DGR, and how the clay behaviour around the EDZ would act. The result of this was that the induced heat from HLW would certainly create perturbations in the clay formation. Such as increasing porewater pressure as shwon in the paper that the generated heat from PRACLAY test that is 80 deegree C are able to heat the clay liner interseaction up until 50 degrees C. Increasing heat are also able to extend and add plastic strain to the already deformed ground due to the excavation.

These findings indicated that increases in temperature correlate with increases in hydraulic conductivity and water pressure. This behavior may be attributed to temperature-induced changes in water viscosity. Higher temperatures also appear to increase the density of the host formation, likely due to the added pressure accompanying temperature rise. Kursten (2024) [3] examined the effects of canister corrosion on DGR stability. It was noted that anoxic corrosion of the canisters can lead to gas buildup within the DGR, causing pressure increases that could eventually result in fractures around the tunnels. The thermo-hydromechanical behavior of Boom Clay in the HADES URL was also studied, with a focus on chemical alterations (Li, 2023) [4]. Findings showed that heat can alter the chemical composition of the host rock, potentially leading to the dissolution of minerals. Such chemical changes may modify the mechanical properties of the rock, impacting the long-term stability of the repository.

Modelling

3.1. Model Geometry

The plan view shows three circular basins arranged horizontally. The leftmost basin is connected to a vertical rectangular basin above it, which is labeled '70m'. The distance between the centers of the first and second circular basins is labeled '60m'. The distance between the centers of the second and third circular basins is also labeled '60m'. The radius of the middle circular basin is labeled '1.6m'. The vertical rectangular basin on the right is labeled '140m'. The horizontal distance from the left edge of the rectangular basin to the center of the first circular basin is labeled '500m'. The vertical distance from the top edge of the rectangular basin to the center of the first circular basin is labeled '430m'. The vertical distance from the bottom edge of the rectangular basin to the center of the first circular basin is labeled '570m'.

The three-tunnel configuration has been selected to observe the interactions of forces and heat generated from each drift, allowing us to assess how each drift affects the others. Additionally, I will generate another model with increased spacing between the tunnels but with thinner concrete walls.

The boundary conditions that I used in this models are that the models is isotropic and fully saturated all the time. Besides that I set the pressure gradient of σ_v is 10kPa/m and the K_0 is 0,9. So the total stress at the center of the tunnel would be 50MPa. I represent the overburden pressure with a 1 m layer of soil with a modified γ to obtain this results.

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free to move in both directions, while the bottom boundary is fixed in all directions.

I also set a temperature boundary condition on every boundaries. The problem states that the initial condition of the clay formation at 500m depth is 295K, thus I set the temperature of the boundary condition to be 295K in all of sides of the boundary.

3.3. Physical Process

As mentioned in the 2.3 we could define there are three main process that occurs on the DGR. First are the mechanical process, thermal, hydraulic and to some extent chemical processes.

Mechanical

The major mechanical process during this construction is the redistribution of stress around the excavated area. As mention before the excavation around the tunnel would introduce EDZ. Where plastic deformations happens around the excavation area. This would then able to cause fracturing which could cause the hydraulic permeability of the rock increase. Allowing then fluid and radiation seeps through it. During this phase I would model a 3 tunnels that separated 60m apart from each other. The first phase are gonna be the excavation phase where the liner has not installed yet then it will follow by the model where the liner are already installed.

Hydraulic

The hydraulic process in DGR mostly came from porewater pressure and existing groundwater if there's any. DGR is used to store radioactive waste. It's normally located on area that does not interact with any groundwater table to prevent any infiltration to any waterbodies that are use by human activity. The porepressure on the rock increased because the fluid dilation is bigger than the solid dilation causing fluid to expand faster than solid. Thus it creates a higher porewater pressure.

Thermal

The thermal process in DGR came from emitted heat from the radioactive waste. The thermal emitted by the waste would then subsequently able to change the mechanical properties of rock and also water that are existed in rock. One of the process that could occur because of the thermal emission is thermal strain. The thermal strain of the rock could cause an increase the rock stress due to the rock expanding. Thermal process could also changes the viscosity thus leading to higher permeability of the fluid.

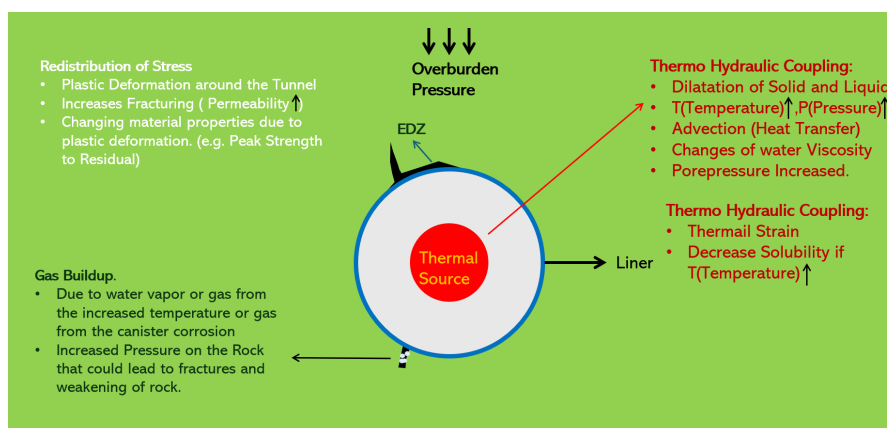


Figure 3.2: Physical Process

Figure 3.2 is the illustration of the physical process that could occur at the tunnels. Here I incorporated all of the physical processes that has been described in previous chapter 3.3. The images is a transversal cross section of tunnel where I incorporated the lining, pressure, and thermal source that affects the model.

3.4. Modelling Material

The material that are used in modelling the DGR are the indurated clay that act as a host formation and concrete that would be use a liner barrier for the DGR. The materials use for modelling could be seen in Appendix A.

3.4.1. Heat Decay

The decay rate of the could be seen at table 3.1 below.

Table 3.1: Heat Decay

Years	Days	W/m	W/m^2
0	0	300	29.84155183
5	1825	278	27.65317136
10	3650	259	25.76320641
15	5475	241	23.9727133
20	7300	226	22.48063571
25	9125	212	21.08802996
30	10950	201	19.99383973
40	14600	180	17.9049311
50	18250	164	16.31338167
60	21900	151	15.02024775
70	25550	140	13.92605752
80	29200	132	13.13028281
90	32850	125	12.43397993
100	36500	119	11.83714889
110	40150	114	11.3397897
120	43800	110	10.94190234
130	47450	106	10.54401498
140	51100	103	10.24559946
150	54750	101	10.04665578
160	58400	98	9.748240264
170	62050	97	9.648768425
180	65700	94	9.350352907
190	69350	92	9.151409228
200	73000	90	8.952465549
210	76650	88	8.75352187
220	80300	87	8.654050031
230	83950	85	8.455106352
330	120450	73	7.261444279
430	156950	62	6.167254045
530	193450	53	5.27200749
630	229950	46	4.575704614
730	266450	40	3.978873577
830	302950	34	3.382042541
930	339450	30	2.984155183

The heat would be simulated using a thermal line that comes from the inside of the tunnel with 50 lines at each tunnel to simulate the heating process due to the radioactive waste.

Excavation Phase

For the excavation phase the main driving forces of it would be the mechanical coupling caused by redistribution of forces due to excavations. Here I would model the tunnel from the construction phase before the liner installment and after the liner installment.

Emplacement Phase

During the operational life of the facility mainly the acting forces would almost be similar with the ones during excavation phase therefore no modelling would take places during emplacement.

Early-Post Closure

The early post-closure are the times where the heat emitted from the repository are at it's maximum but the temperatures around the tunnel has not build up yet. In this model I classified the early-post closure in 100 and 200 days after the closures

Late Post Closure

In the late post-closure phase, investigations would focus on how heat buildup affects the temperature, behavior of the surrounding rock formations, and the stability of the underground facility. This phase examines the long-term thermal impacts on the geological environment and assesses any potential alterations in rock properties or stress distributions. I classified the late post closure in 20.000 days until 320.000 days.

4

Analysis

In this chapter I would look upon the changes that happen with DGR construction for the surrounding formation and the tunnel. The parameters that I would like to find out is what is distances between the tunnel and also the thickness of the liner.

During this analysis I would create 2 models which has 2 different parameter. First model has 60 m spacing apart between each tunnel and a 0.7 m liner thickness while the second model has 100 m spacing and 0.4m liner thickness. To compare results across different phases and parameters, I chose a cross-section that includes all three tunnels I've created horizontally. For consistency in comparison, this cross-section remains the same across all graphs, except when contour plots are more suitable for illustrating a behavior during a specific phases. The defined cross section could be seen below where the A cross sections covers all three tunnels horizontally and cross section B cover tunnel in the middle which we would state as tunnel 2 vertically.

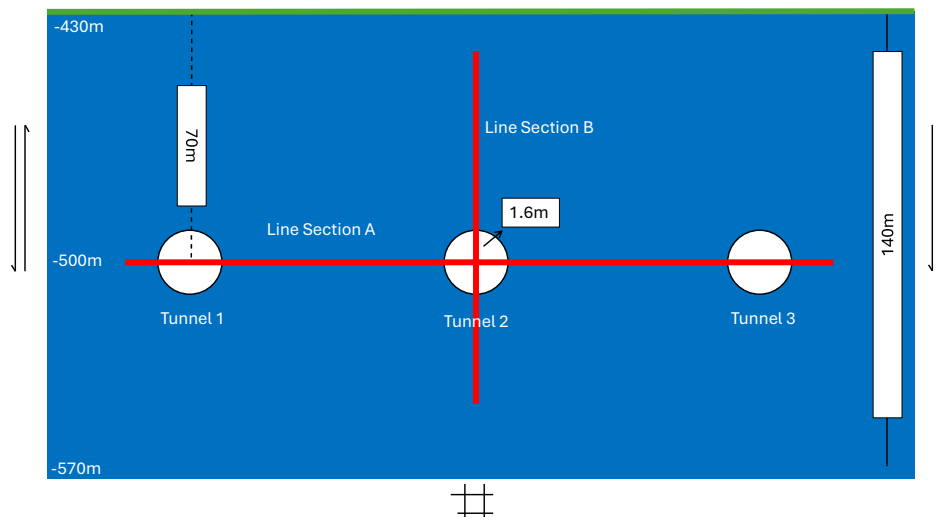


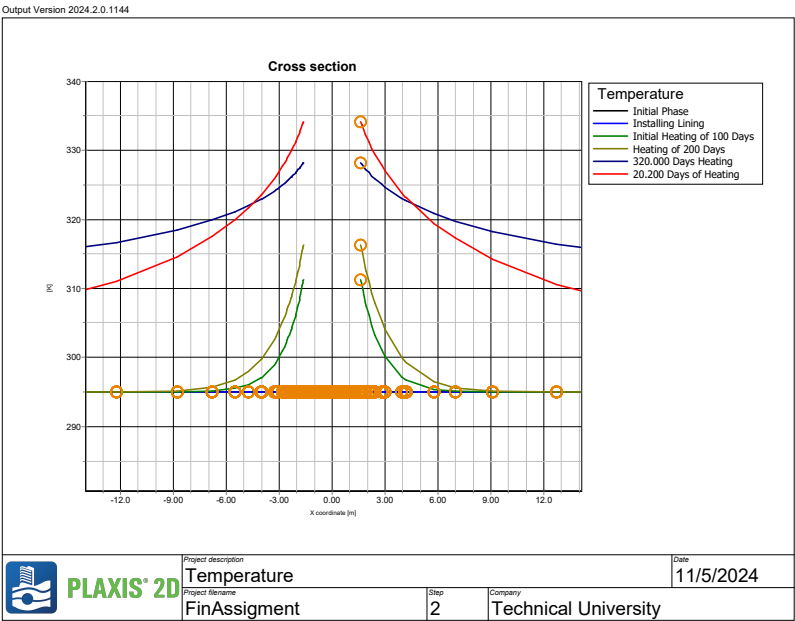
Figure 4.1: Cross Section

4.1. 1st Case

In the first case, the tunnels are spaced 60 meters apart from center to center, with each liner having a thickness of 0.7 meters. To compare results across different phases and parameters, I chose a cross-section that includes all three tunnels I’ve created. For consistency in comparison, this cross-section remains the same across all graphs, except when contour plots are more suitable for illustrating parameter behavior during specific phases. Below, I present results from implementing the Deep Geological Repository (DGR) model with the 60 m spacing and liner thickness of 0.7.

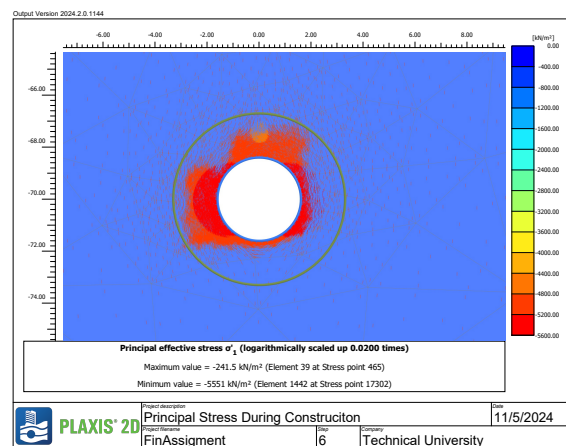
4.1.1. Temperature Changes

The temperature changes is shown below. During the early emplacement phase the surrounding temperature increase from 295K to 310K and further reaching 330K after 930 years.

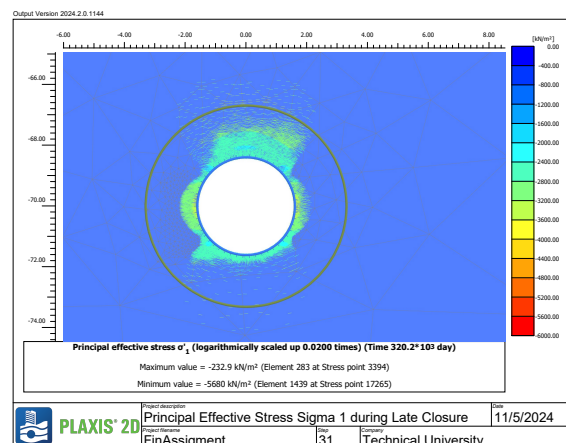


From the cross section A, it could be observed that as time goes on the temperature of the surrounding tunnel kept increasing. The temperature closest to the tunnel are the highest. The interesting part are during the 20.100 aays we could observe that the temperature near the tunnel were at the high-est. Those are the time when the heat emitted by the radioactive waste got halved than the initial value as shown in table 3.1. Since that point the temperature near the tunnel keeps decreasing while temperature around the host formations still builds up and increase up until 310 K.

4.1.2. Stress

Figure 4.2: Principal Effective Stress σ_1 During Construction

From the contour images Figure 4.2 we could observe the difference of the effective principal stress of σ_1 . The introduction of tunnels changes the σ_1 that was once vertical due to the overburden pressure to be radials because of the excavation. The value of stress increases around the tunnels increase around 6 MPa. Before the excavations the σ_1 is 50 MPa. Once heating is introduced the σ_1 change at a small rate but during the late closure we could see that the principals effective stress σ_1 completely changes to be horizontal and to lower value in 2,8 MPa as seen in the figures.

Figure 4.3: Principal Effective Stress σ_1 During Late Closure

This changes of stress direction and magnitude are due to the expansion that was caused by the built-up of the temperature over a long time. The reduced stress is caused by the tensile stress occurring at the tunnel during the thermal expansion.

4.1.3. Deformation

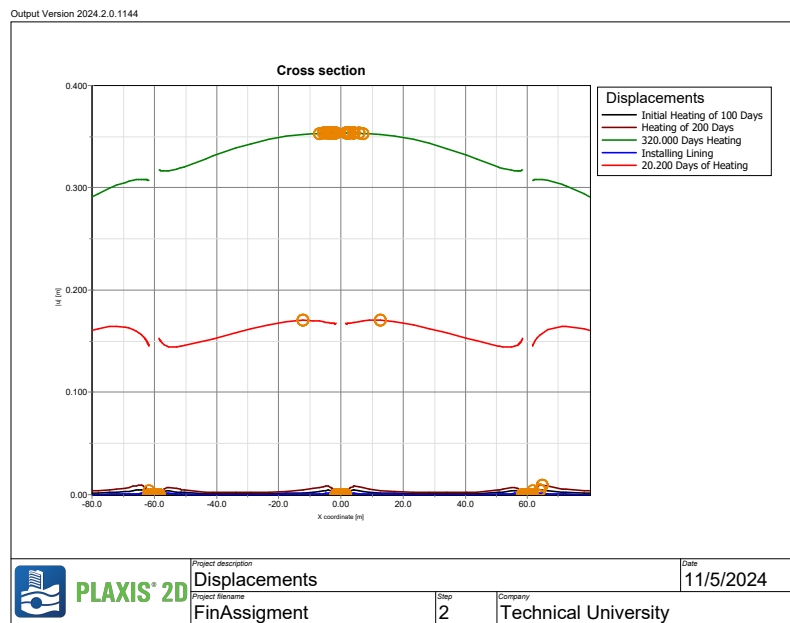


Figure 4.4: Displacements

We could see in the figure 4.4 that the deformation during the construction and even early heating phase are very small compared to the long term closure displacements. It shows that here the biggest displacements occurs during the late closure phase due to the heating of the surrounding rock. The deformation during the construction are around 0.001m while at the long term it could grow to 0.3m displacement.

4.1.4. Porewater Pressure

Typically, in tunnel projects, the porewater pressure around the tunnel is reduced to zero due to drainage implemented during construction. However, in this scenario, we assume that the surrounding formations remain saturated throughout. As a result, the excavation process would actually lead to an increase in porewater pressure around the tunnel as we could observe from figures 4.5 .

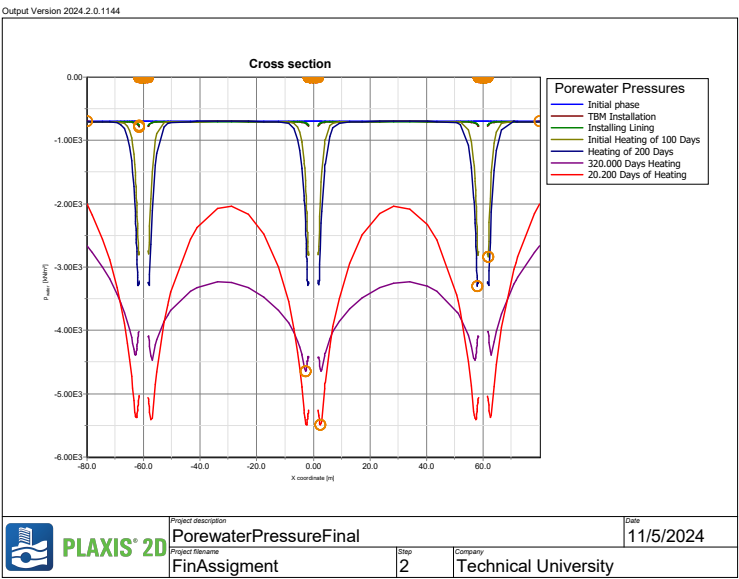


Figure 4.5: Porewater Pressure

The pore water pressure at the early post closure is significantly bigger than during the construction phase. The increase of porewater pressure is caused by the dilation of fluids and it's molecular activity that has been explained in the section 3.3. The values rise up from 760 kPa in the construction phase to 2.000 kPa during the early closure phase. And doubling of the values from the 2.000 kPa to 5.000 kPa near the tunnels during long-term closures.

4.1.5. Volumetric Strain

The appearance of the underground excavation cause a shrinking in the surrounding tunnel and host formation which here shown by the negative strain during the construction phase. Due to the small size of the tunnel the volumetric strain also has a small value. The value of ϵ_v is around - 0,001 during the construction phase.

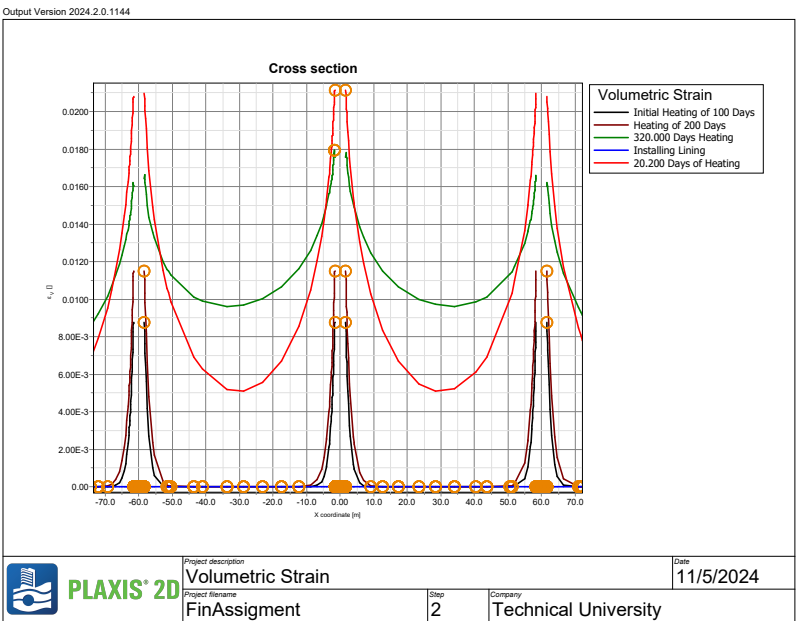


Figure 4.6: Volumetric Strain

However during the initial heating the figure 4.12 shows that in comparison during the excavation phase the strain are positive. Which mean the effecting temperature cause a expansion upon the soil instead of compression. The values of the volumetric strain in the early closure phase are also way bigger compared to the construction phase. We could observe that during the late closure the value doubled. Reaching a 0.02 volumetric strain at near the tunnel.

4.2. 2nd Case

In the second case, the tunnels are spaced 100 meters apart from center to center, with each liner having a thickness of 0.4 meters. The 100 meters spacing are chosen because the (Li,2023) [4] shows that for a HLW a spacing of 100 - 120m is recommended.

4.2.1. Temperature Changes

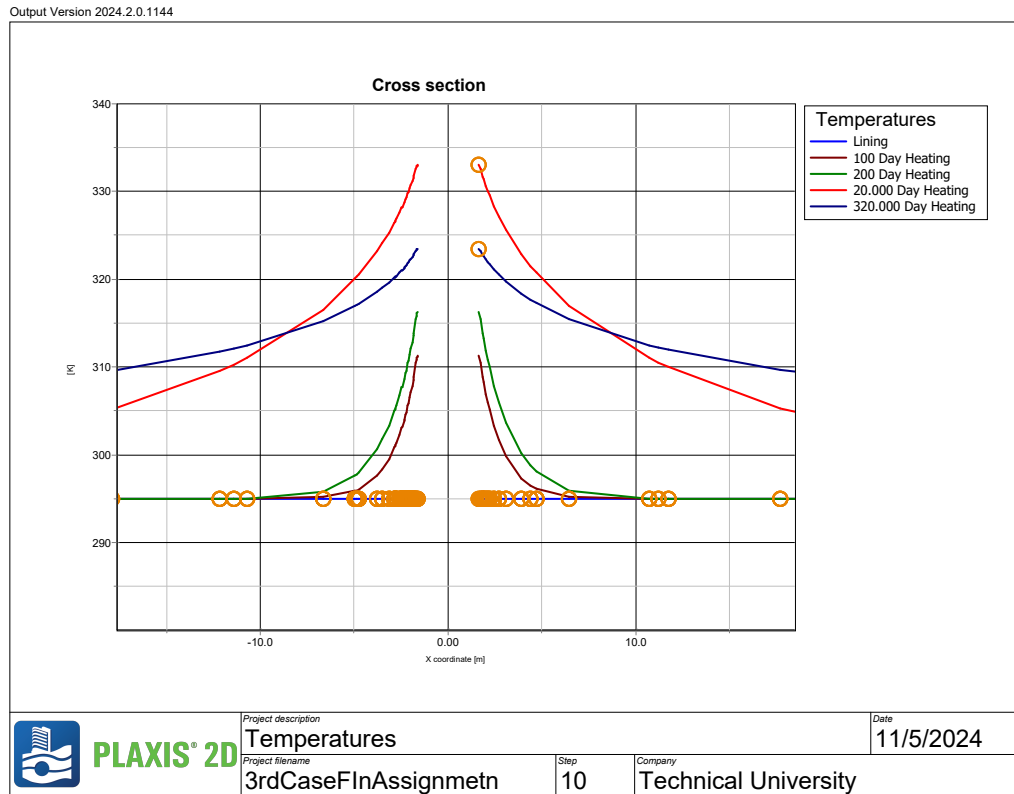


Figure 4.7: Temperature Through Various Phases in 2nd Case

The temperature around the tunnel does not change much as the emitted heat has the same value. However, the temperature between each tunnel shows a significant reduction in range of 10K. This is due to more spacing which led the emitted heat to have less interference with the other source that comes from another tunnel. Which will in turn reduce the volumetric strain and displacement between the host formations that is located between the tunnels.

4.2.2. Stress

The stress obtained at the tunnel using the second model also shows similar value and behaviour where first the tunnel is dominated by compressive stress due to overburden pressure then overtime the tensile stress would eventually have an effect due to the thermal strain induced by temperatures on the rock. Here we could observe that the σ'_1 go from 56 MPa during construction phase to 20 MPa at the late closure stage. The principal stress direction also changes completely due to the forces induced by the heat.

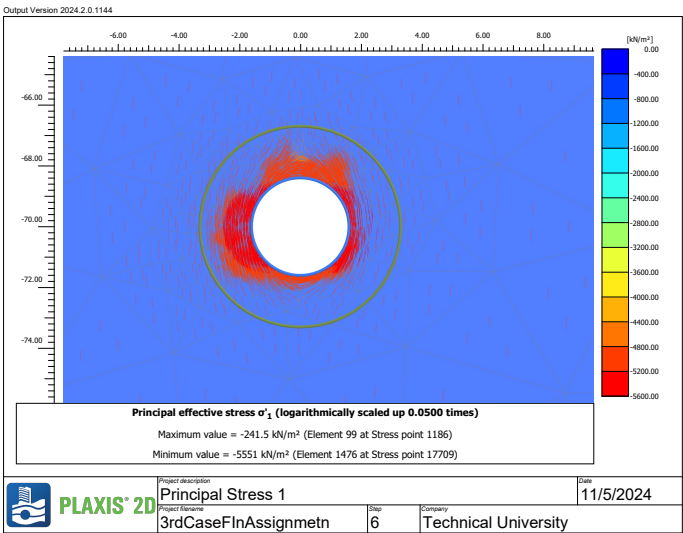


Figure 4.8: Effective σ'_1 Construction Stress

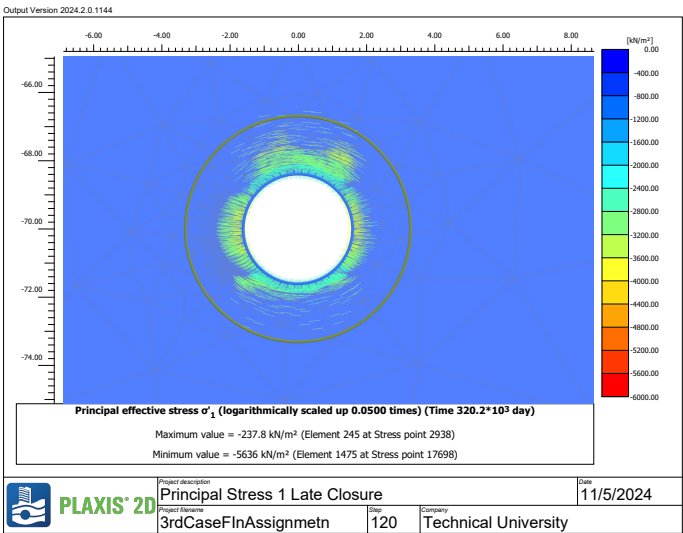


Figure 4.9: Effective σ'_1 Late Closure

4.2.3. Deformation

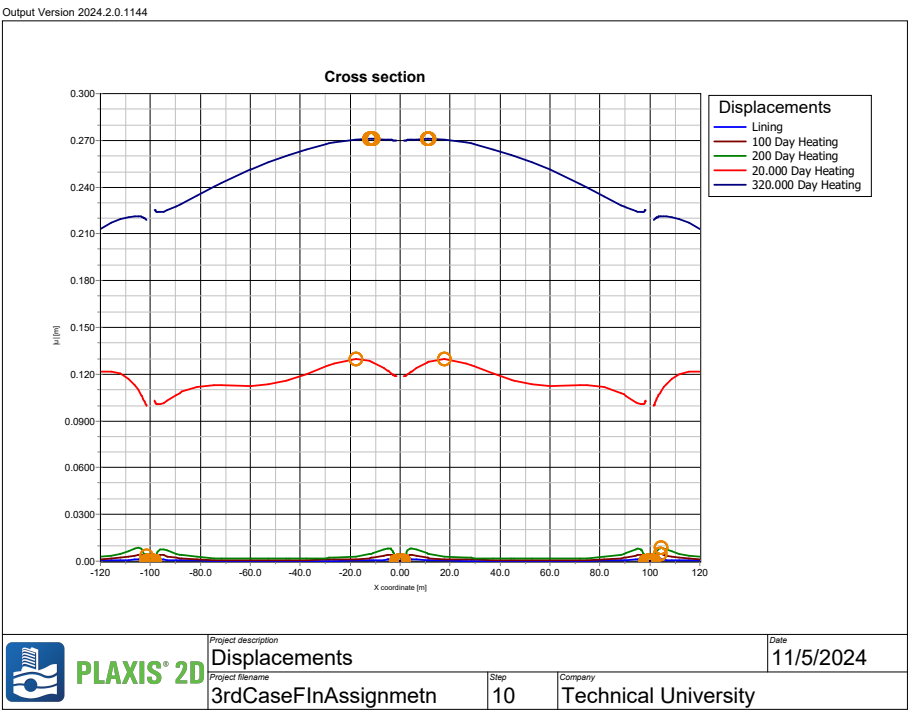


Figure 4.10: Total Displacements

The deformation of the second phase could be seen in Figure 4.10. The value during the early construction changed a bit due to the different parameters. The crown deform to 4.5 mm in comparison of 5 mm to the 1st case. Thinner liner would actually cause the deformation at the tunnel to be bigger due to forces of the concrete to it's thinner. However, an overall view of the boundary indicates that the 100 m spacing between tunnels prevents any significant interference in the force distribution thus the u_y of the tunnel with the second are smaller by a bit. But a big difference in displacements is observed during the late closures. The first model shows a displacement around 0.33 m at the tunnel 2 while the second model shows a displacement at only around 0.27 m. This is due to the spacing that creates less interference of the heat buildup and therefore reduce the displacements that occurs on the area.

4.2.4. Porewater Pressure

The porewater pressures near the tunnel observe the same behaviour with the first case although smaller. p_{water} of 750 kN/m^2 at maximum near the tunnel. Results could be seen in Figure 4.11.

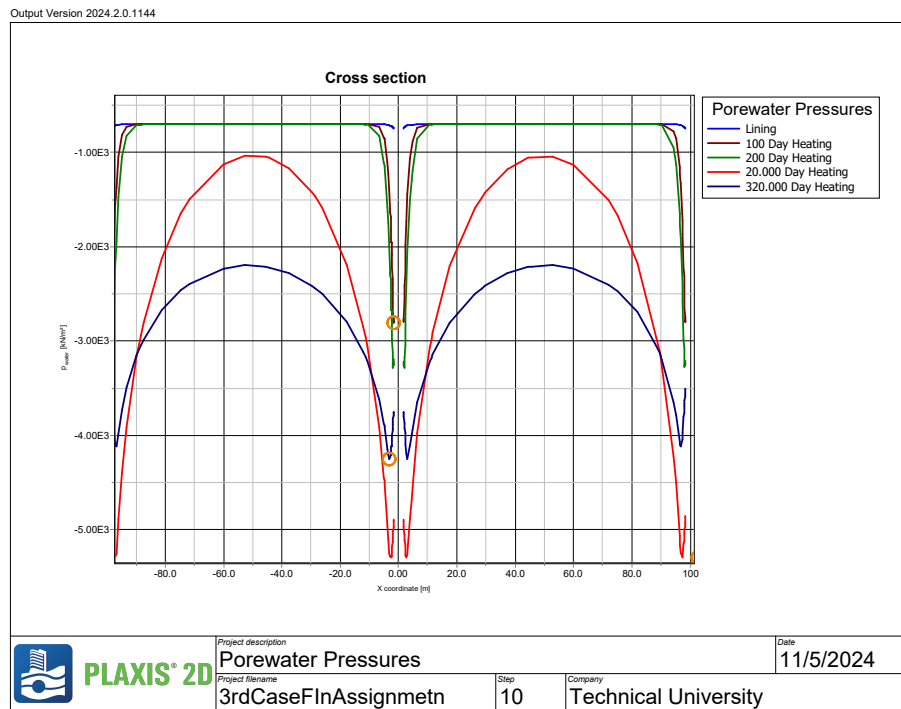


Figure 4.11: Porewater Pressures

Pore-water pressure p_{water} would then increase significantly just like the first models. But here we could see that in between the tunnels the pore-water pressure are much smaller in the second model. Again this is caused by the spacing of second model that is larger which cause less heat and temperature buildup that reduce the porewater pressures as the results.

4.2.5. Strain

The volumetric strain near the tunnel observe the same behaviour with the first case with almost similar value around ϵ_v of -0.0160×10^{-3} at maximum near the tunnel. The negative convention shows that the tunnel is shrinking due to excavation.

The ϵ_v around the tunnels shows a similar value in this phase with the first model. The ϵ_v is 0.021 at 320.000 days of heating. But we could see a significant reducement of the volumetric strain at the rock between the tunnels. This is caused by the less temperature that build up between those tunnel.

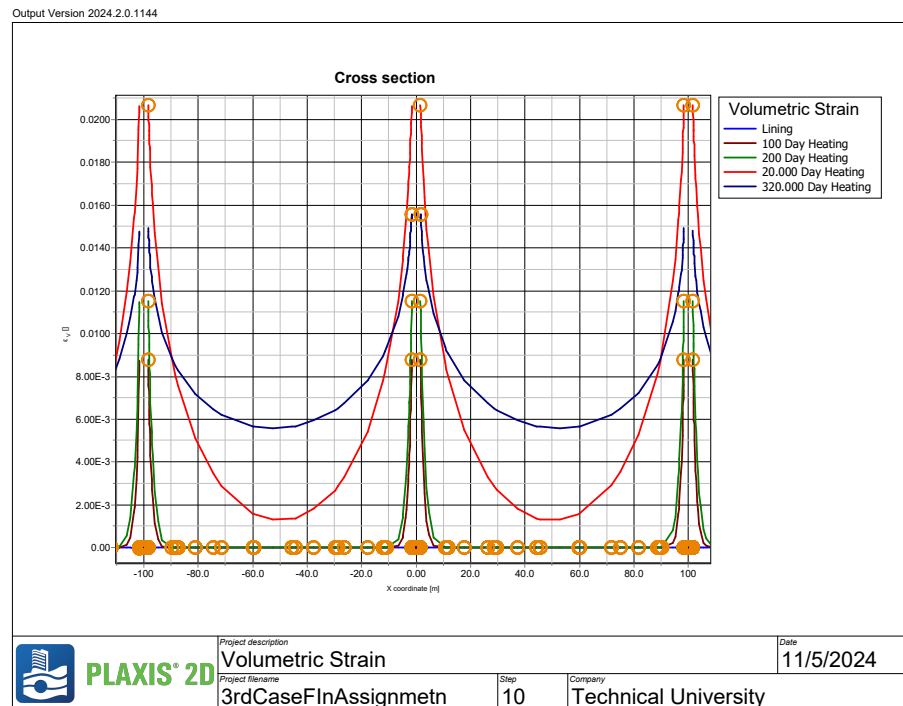


Figure 4.12: Volumetric Strain

4.3. Comparison between 1st and 2nd Case

The tunnel shows all a similar value and behavior during the construction and early closures between these model. This is due to to the same geometry and same forces that works in the tunnel. One interesting notes is that the second model obtain less deformation in a thinner lining. This in contrast to the usual condition when thicker lining would produce less deformation. This behavior could be caused because the first model has less distance between each tunnel which lead to a distribution of force between each tunnel interfere more with each other.

Big difference are shown with the rock that are presents between the tunnels especially during the late closures. As the tunnels has more spacing it would causes less interference of the heat which in turn reduces the thermal expansion than a chain effect that would also reduces the pore-water, tensile stress, displacements and also temperature of the rock between them. From here we can concludes that the spacing plays more role towards how the radioactive waste affects the surrounding rather than the liner thickness.

4.4. Recommendation

The primary forces affecting the deep geological repository (DGR) during long-term closure are temperature and the tensile stress generated by temperature buildup. The small tunnel radius does not introduce significant forces during the construction phase and has a minimal impact on stability. Therefore, it is recommended to use reinforced concrete for the tunnel liner, as it can better withstand tensile stress compared to normal concrete. While also adhere to material that could withstand the heat generated by the waste. Additionally, maintaining sufficient spacing between tunnels will help minimize heat buildup, which in turn can reduce pore water pressure, tensile stress, and deformation resulting from the generated heat at the long-term closure phase.

5

Conclusion

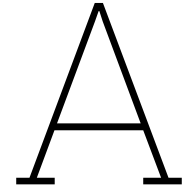
The construction of the tunnel induces a redistribution of forces and alters the stress field, which subsequently leads to changes in strain, deformation, and the direction of principal stress. Additionally, the heat generated by the radioactive waste will dominate the changes occurring near the deep geological repository (DGR). Over the post-closure period, temperatures are expected to increase from 295 K to 330 K at maximum, particularly around the tunnel, due to thermal buildup over approximately 930 years. This rise in temperature introduces thermal strain in the surrounding area, consequently increasing the tensile stress, deformation, strain and pore-water pressure.

It shown in this research that bigger deformation occurs across the host formation during the late phase closure, the thermal forces produces expansion of the rock which in turn cause a tensile stress on the construction. Complimented by the increasing value of the volumetric strain and also the changes of principal stress due to the introduction of the thermal process.

The second case models the DGR with more spacing of 100m apart from the tunnels and with lining of thickness 0.4m. The second models shown less deformation although having a thinner lining. But with a spacing of 100m it could be observed that the surrounding has less interference on the temperature and the redistribution of force from the tunnels. The temperature near the tunnel are almost the same becaus both model has the same heat source and radius of tunnel but the surrounding tunnels has less temperature, therefore less volumetric strain and tensile stress on around the host formation.

References

- [1] International Atomic Energy Agency. “Management of Radioactive Waste”. In: (2020). Accessed: 2024-11-05. URL: <https://www.iaea.org/topics/radioactive-waste>.
- [2] Nuclear Energy Agency. “Radioactive Waste Management: Deep Geological Repositories”. In: (2020). Accessed: 2024-11-05. URL: <https://www.oecd-neo.org>.
- [3] Bruno Kursten et al. “Lessons learned from in situ corrosion experiments in the HADES URL”. In: *SCK CEN, Belgian Nuclear Research Centre* (2024).
- [4] Xiangling Li et al. “Forty years of investigation into the thermo-hydromechanical behaviour of Boom Clay in the HADES URL”. In: *EURIDICE, European Underground Research Infrastructure for Disposal of Nuclear Waste in Clay Environment* (2023).



Appendix A Material Properties and Model Set-Up

Table A.1: Indurated Clay Parameter

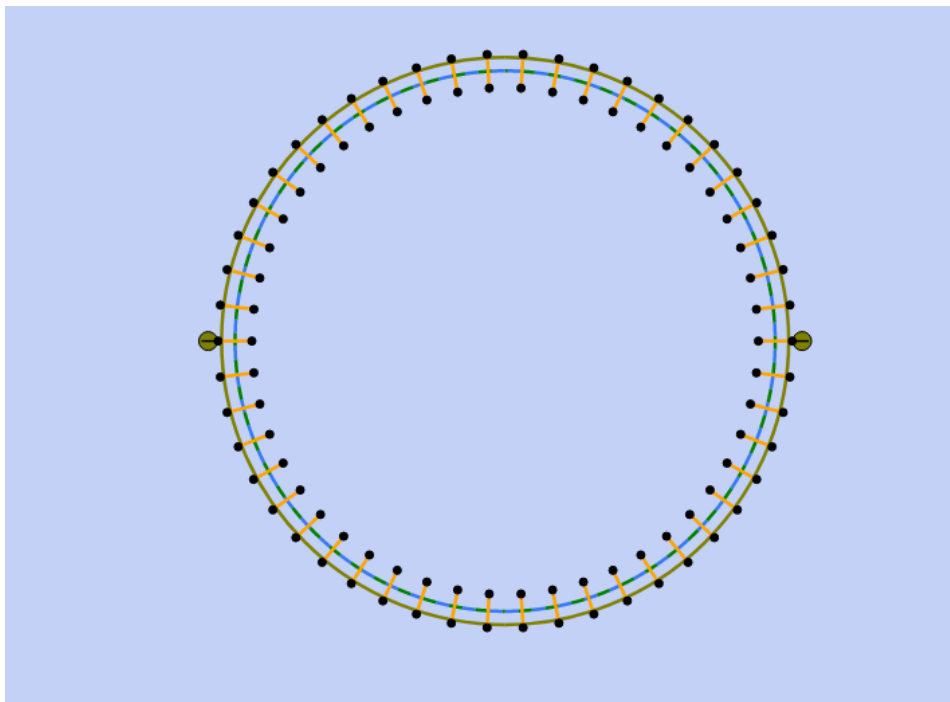
Soil Parameters (Mohr-Coulomb)		
Parameter	Value	Unit
Initial porosity	0.4	-
Young's modulus	300	MPa
Poisson's ratio	0.3	-
Effective friction angle	13	°
Cohesion	0.5	MPa
Hydraulic conductivity	4×10^{-12}	m/s
Volumetric thermal expansion coeff.	10^{-5}	/K
Thermal conductivity	1.4	W/m·K
Specific heat capacity	1400	J/t·K

Table A.2: Liner Parameter 1st Case

Liner Parameters (Elastic)		
Parameter	Value	Unit
Unit Weight	24	kN/m/m
Young's modulus	35000	MPa
EA	300.2×10^6	kPa
EI	36.82×10^6	kPa
Volumetric thermal expansion coeff.	10^{-5}	/K
Thermal conductivity	1.4	W/m·K
Specific heat capacity	8400	J/t·K

Table A.3: Liner Parameter 2nd Case

Liner Parameters (Elastic)		
Parameter	Value	Unit
Unit Weight	24	kN/m/m
Young's modulus	35000	MPa
EA	2.03×10^8	kPa
EI	2.21×10^7	kPa
Volumetric thermal expansion coeff.	10^{-5}	/K
Thermal conductivity	1.4	W/m·K
Specific heat capacity	8400	J/t·K

**Figure A.1:** Simulation of Heat as a Thermal Line in PLAXIS 2D

B

1st Case Results

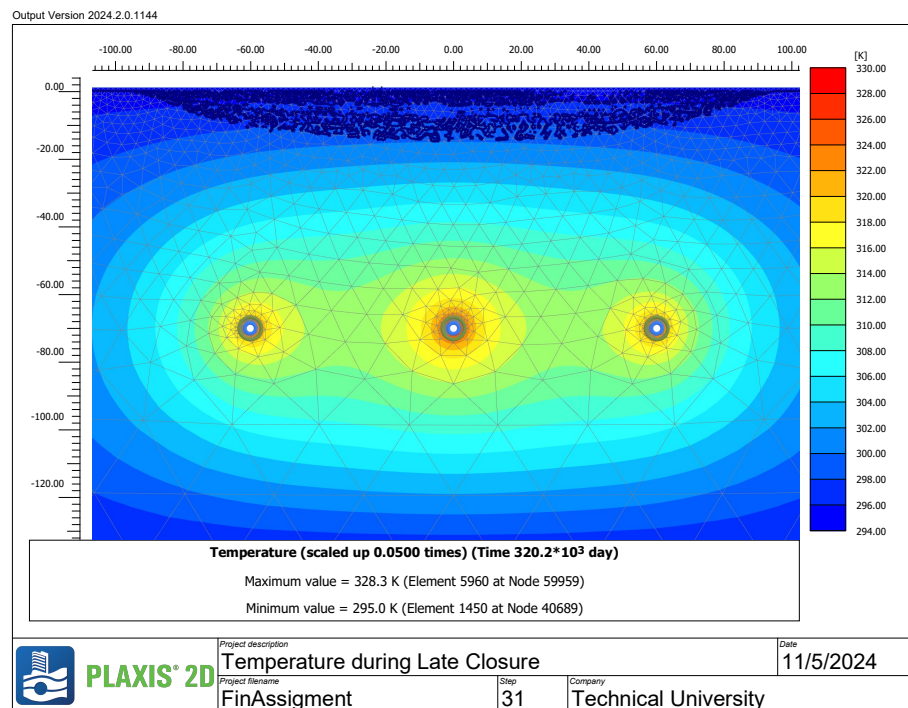


Figure B.1: Temperature Contour at 320.000 Days

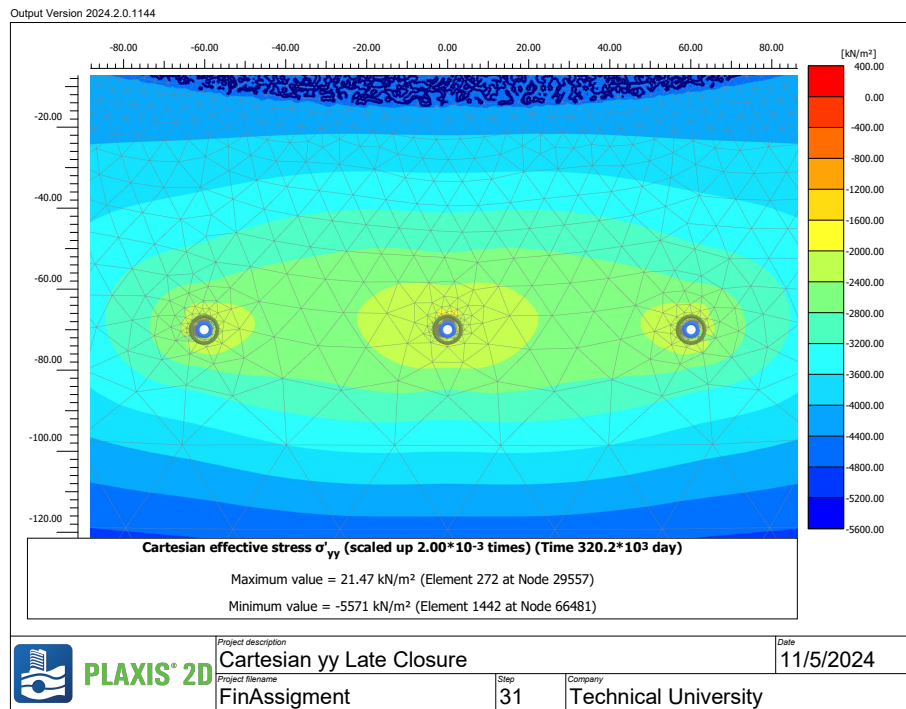


Figure B.2: Cartesian σ_{yy} Contour

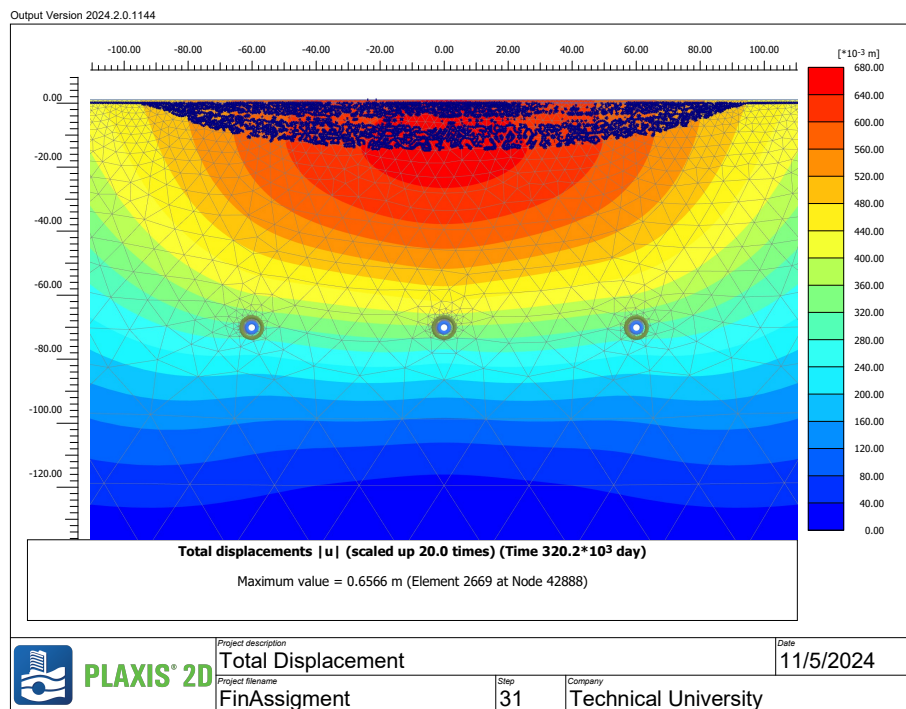


Figure B.3: Total Displacement Contour

C

2nd Case Results

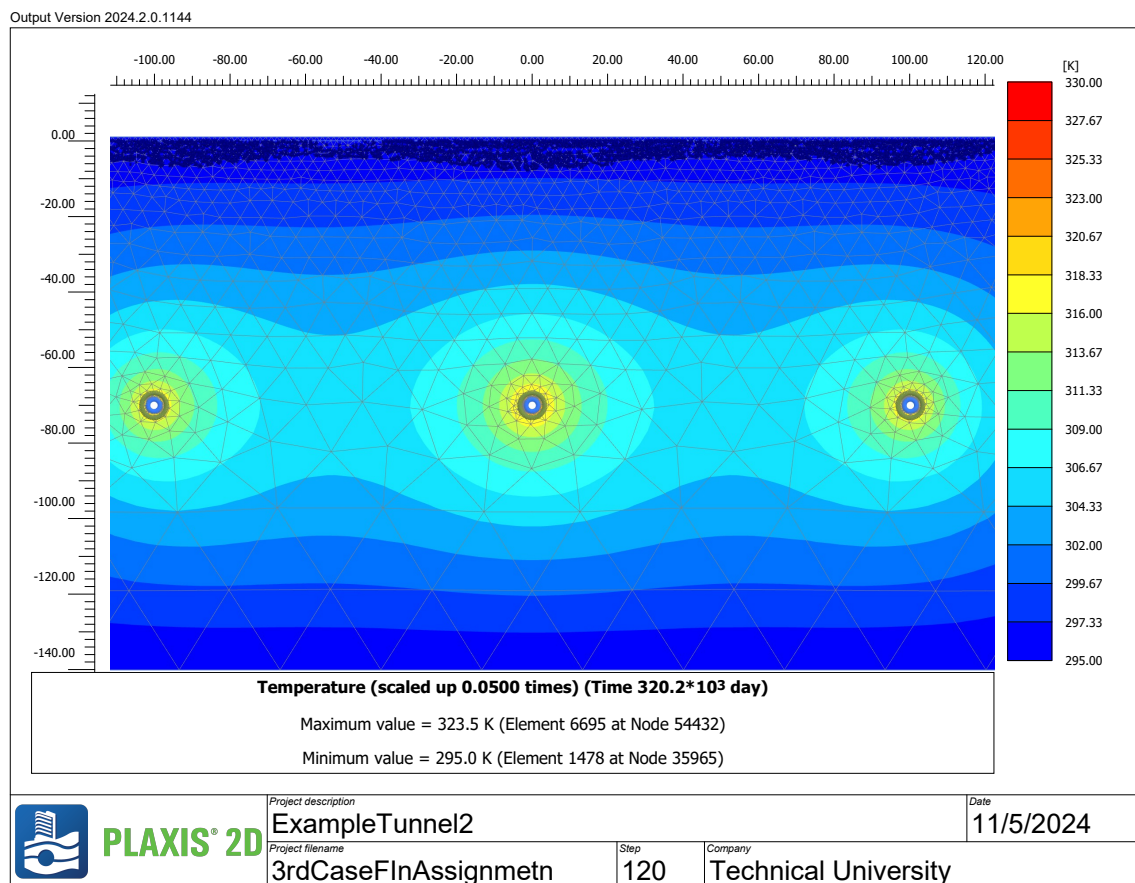


Figure C.1: Temperature Contour at Late Closure

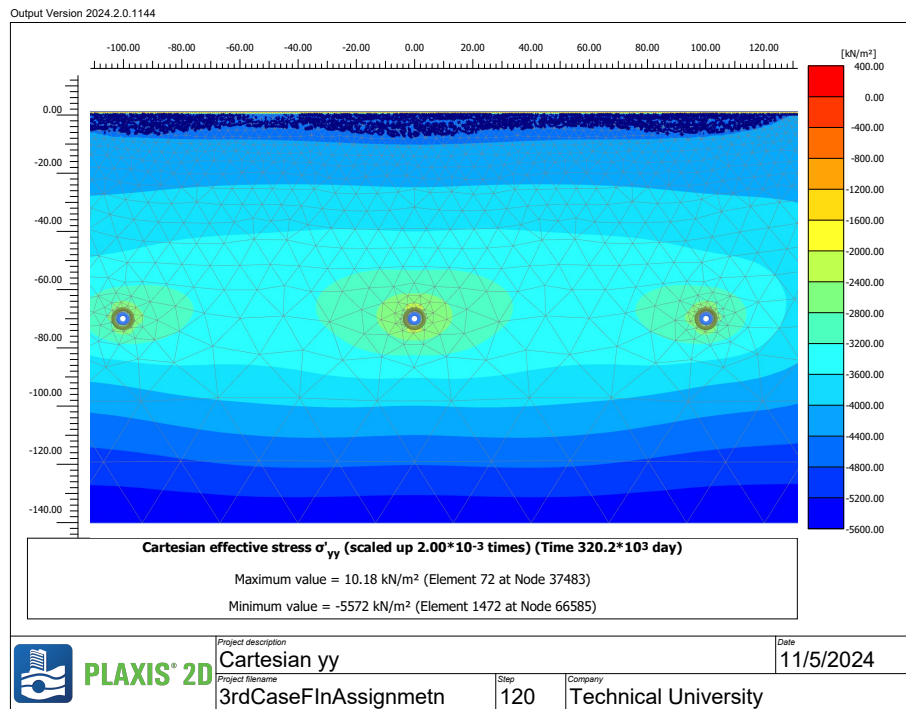


Figure C.2: Cartesian σ_{yy} contour at Late Closure

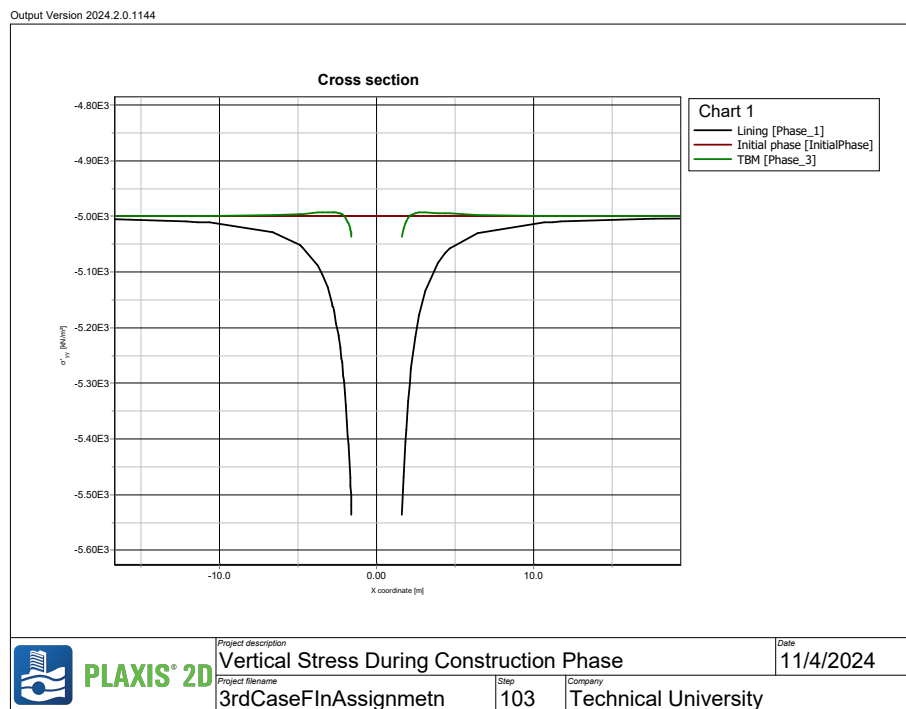


Figure C.3: Vertical Stress during Construction Phase 2nd Model

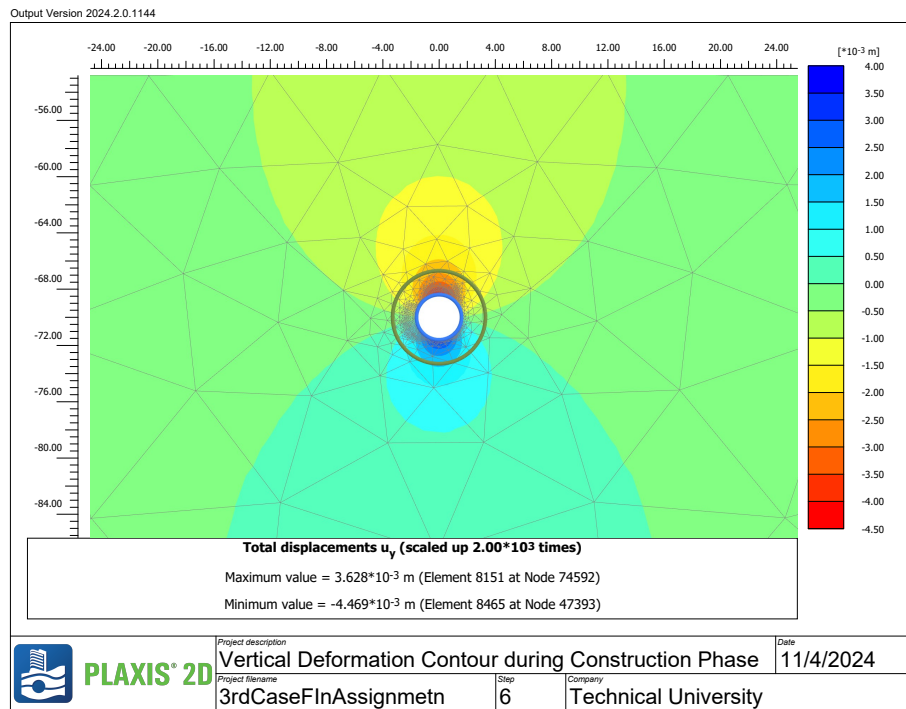


Figure C.4: Vertical Deformation Contour during Construction 2nd Model

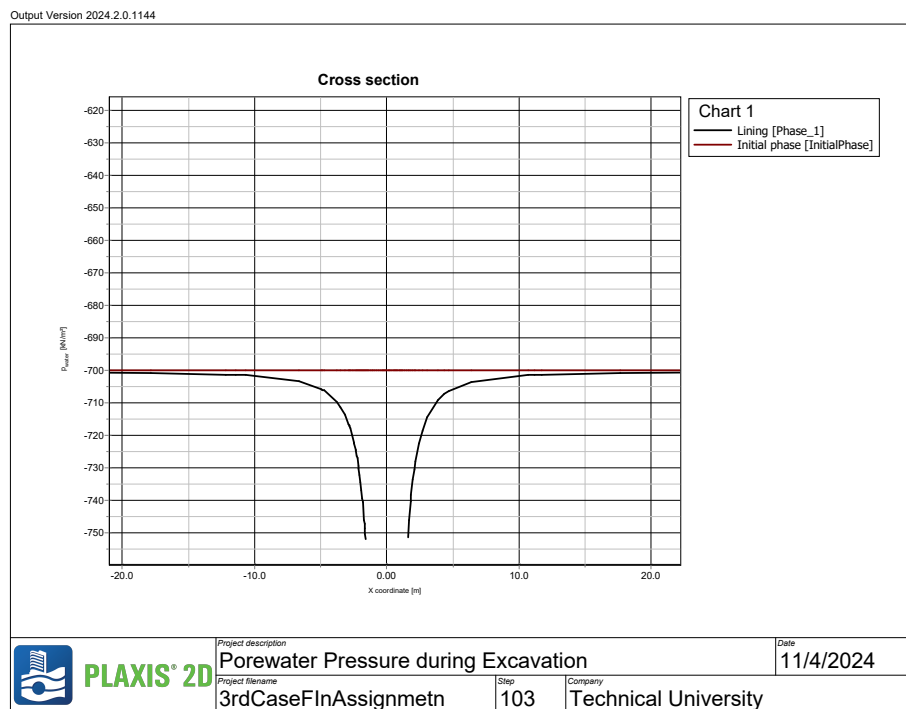


Figure C.5: Porewater during Construction 2nd Model

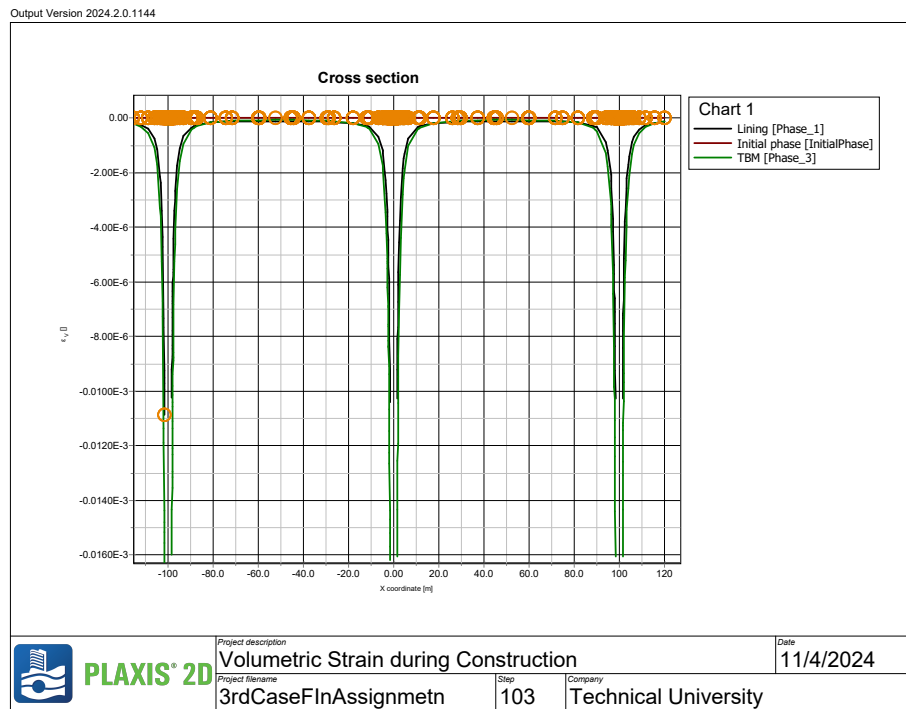


Figure C.6: Volumetric Strain during Construcion 2nd Model

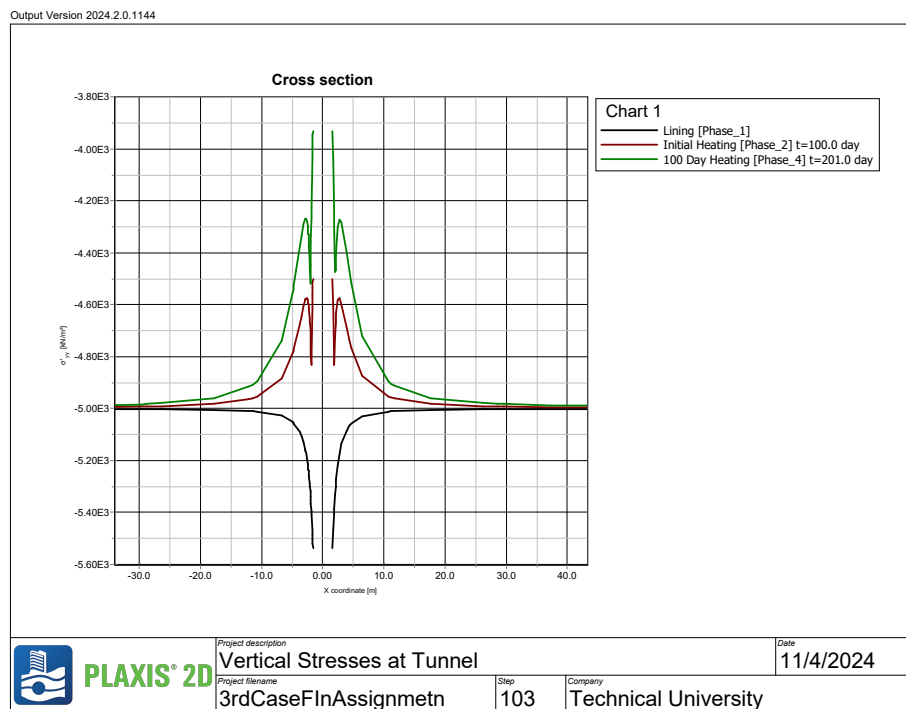


Figure C.7: Vertical Stress at Tunnel during Early Closure 2nd Model

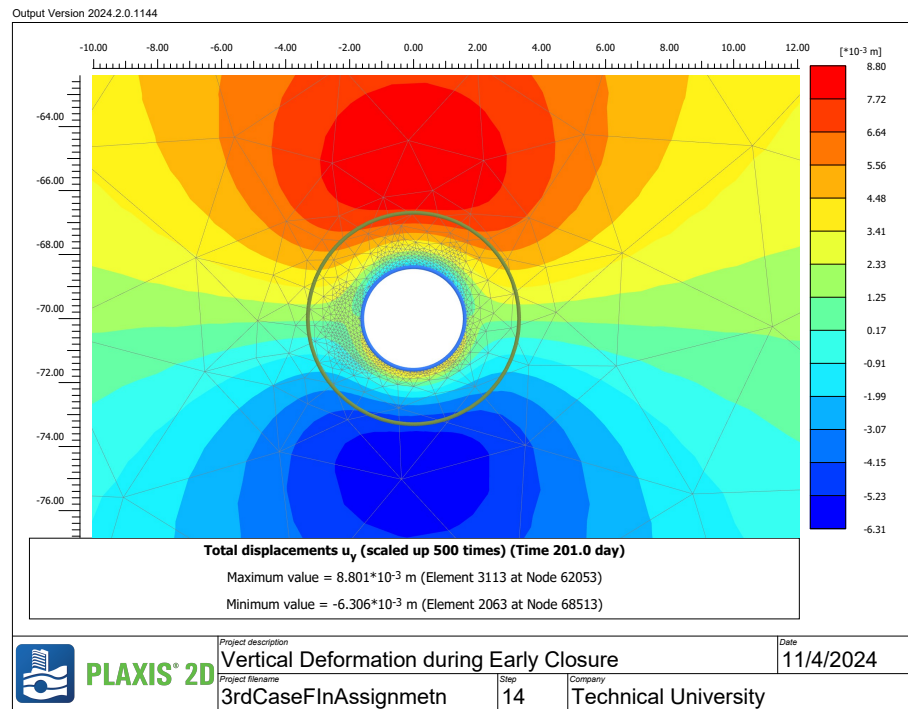


Figure C.8: Vertical Deformation Contour during Early Closure 2nd Model